

Lecture 20. Probabilistic Graphical Models

COMP90051 Statistical Machine Learning

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THE UNIVERSITY OF
MELBOURNE

based on slides by Tommi Jaakkola, MIT CSAIL

Today's topics

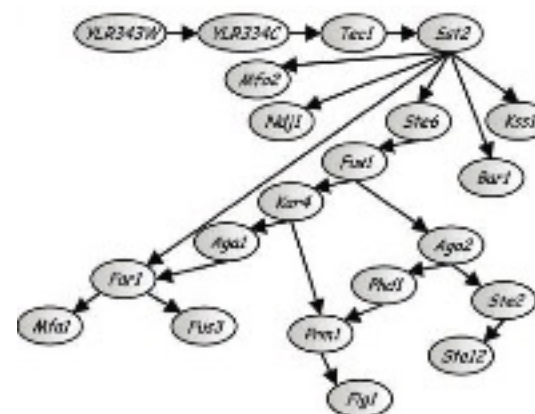
- Bayesian networks
 - * Independence
 - * D-separation
- Markov random fields
 - * Independence
 - * Mapping Bayesian networks to Markov random fields

Bayesian networks

Independence and D-separation

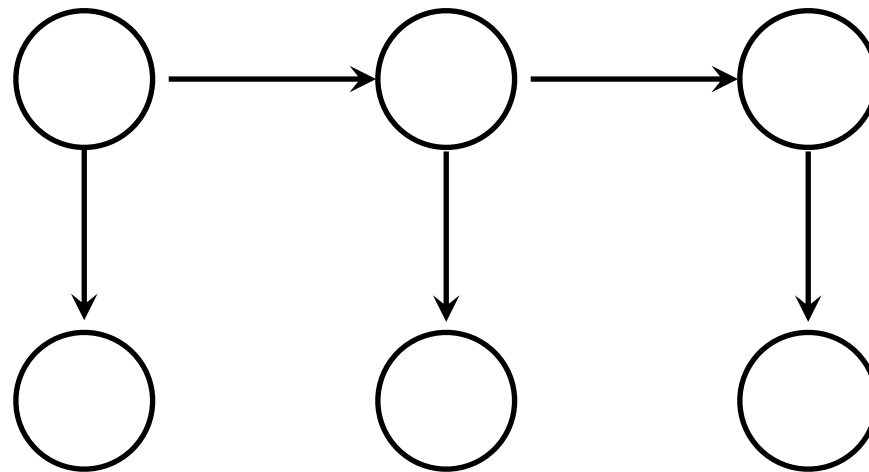
Bayesian networks

- Bayesian networks provide a simple language for succinctly expressing, using, and learning probabilistic information
- There are two parts to any Bayesian network: 1) the graph that explicates how variables relate to each other, and 2) the associated probability distribution that quantifies numerically how the variables depend on each other
- The models are widely used in practice
 - diagnostic inference tools
 - discovery tools (e.g., medicine, computational biology)
 - user modeling
 - etc.

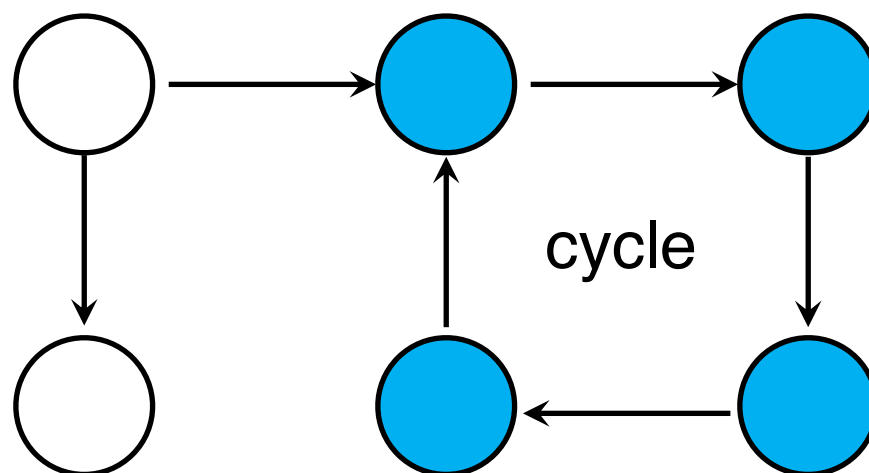


Directed graphs

- A directed graph contains nodes (circles) and directed edges (arrows from one node to another node)

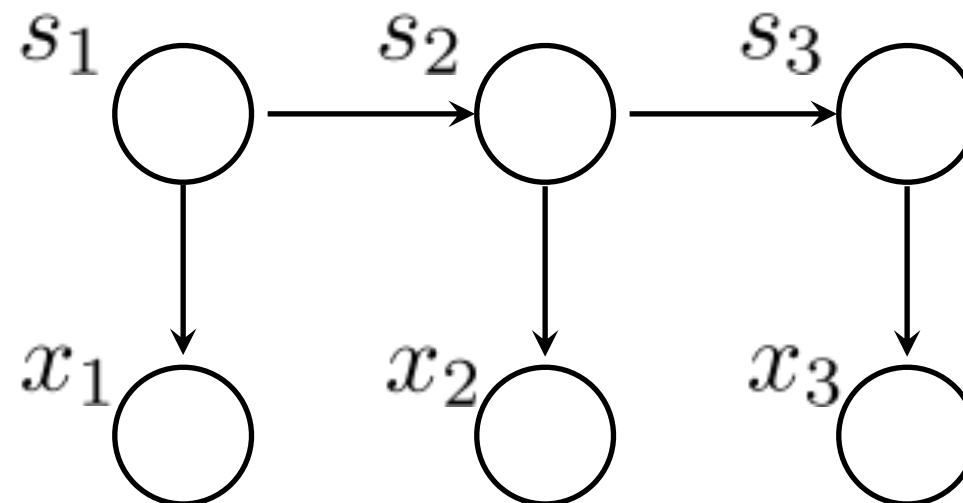


- Assume we traverse nodes by following the direction of the edges. If we start at some node and we end up in the same initial node, we say that there is a cycle



Bayesian networks

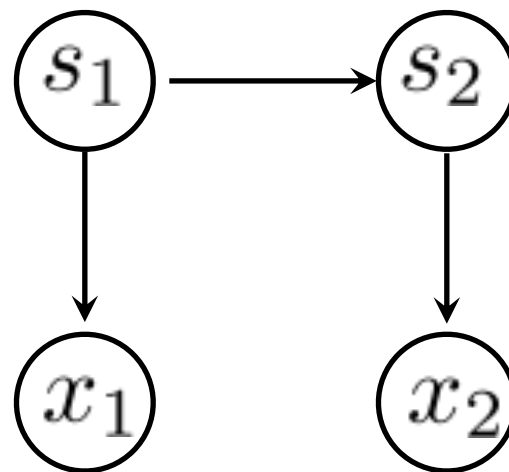
- A directed acyclic graph (DAG) is a directed graph without cycles



- Bayesian networks as probability models are represented by directed acyclic graphs (DAGs) where the nodes specify variables and the directed edges identify dependencies
- Formally, the graph encodes conditional independence properties about the variables (cf. Markov properties)
- Any probability distribution we associate with the graph has to be consistent with such independence properties

Bayesian networks: terminology

- Some basic terminology:
 - s_1 is a “parent” of x_1
 - s_1 and s_2 are “ancestors” of x_2 ; x_1 is not.
 - x_1 is a “child” of s_1
 - x_1 , s_2 and x_2 are all “descendants” of s_1



Bayesian networks: example

- Consider a simple problem with two people flipping unbiased coins independently of each other. Let the variables specifying the outcomes be $x_1 \in \{0, 1\}$ and $x_2 \in \{0, 1\}$.
- A third person reports whether the two coins land the same way ($x_3 = 1$) or differently ($x_3 = 0$). This person reports correctly 90% of the time

Bayesian networks: example

- The two coins are flipped independently of each other

$$P(x_1) : \begin{array}{l} x_1 = 0 \\ x_1 = 1 \end{array} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \quad P(x_2) : \begin{array}{l} x_2 = 0 \\ x_2 = 1 \end{array} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$



Bayesian networks: example

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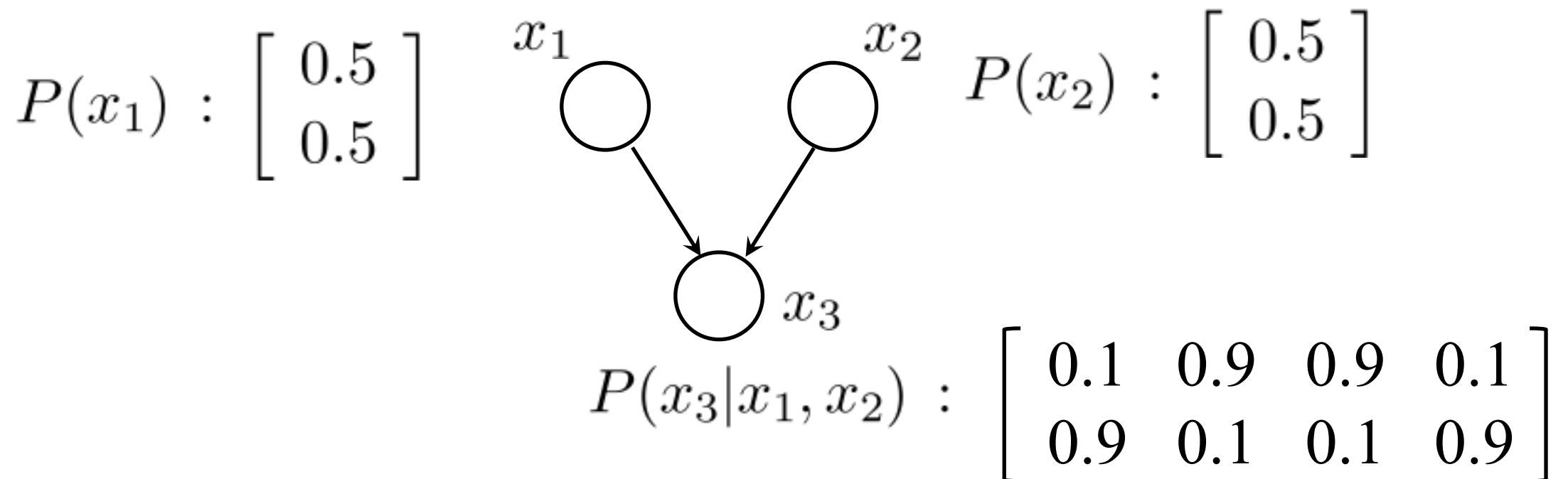
↑
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sum = 1
sum = 1

- The outcome of the comparison is a function of x_1 and x_2

$$P(x_3|x_1, x_2) : \begin{array}{l} x_3 = 0 \\ x_3 = 1 \end{array} \begin{array}{cccc} x_1x_2 = 00 & x_1x_2 = 01 & x_1x_2 = 10 & x_1x_2 = 11 \\ \begin{bmatrix} 0.1 & 0.9 & 0.9 & 0.1 \\ 0.9 & 0.1 & 0.1 & 0.9 \end{bmatrix} \end{array}$$

↑
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sum = 1
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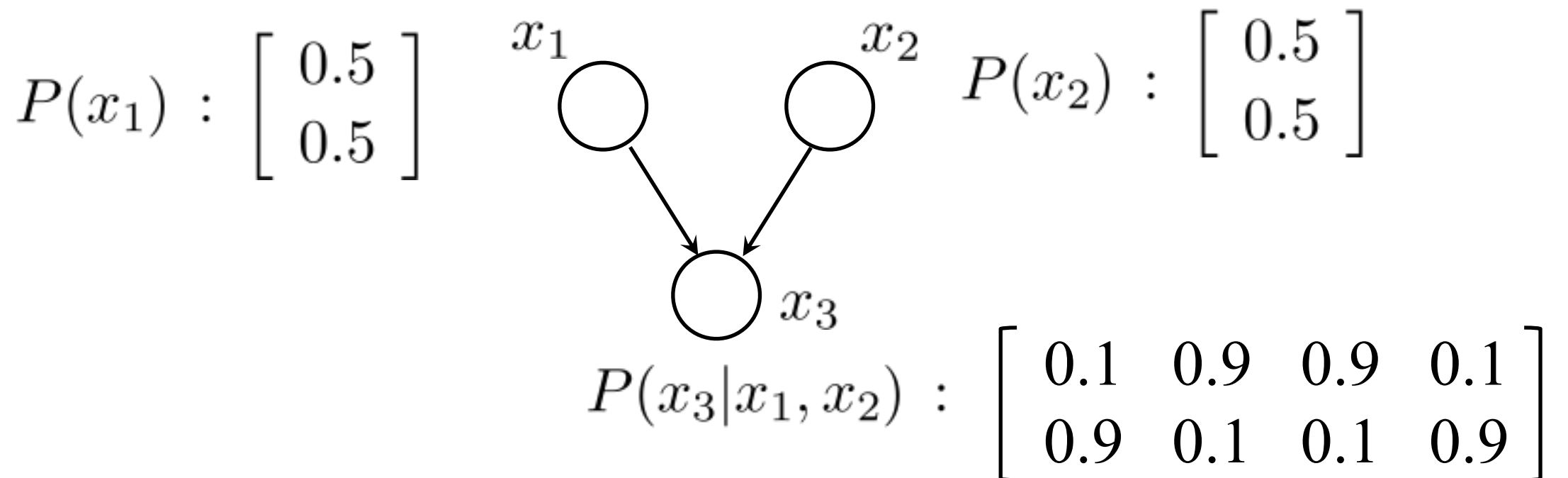
Bayesian networks: example



$$P(x_1, x_2, x_3) = P(x_1)P(x_2)P(x_3|x_1, x_2)$$

- The graph alone would permit us to infer two qualitative properties about the underlying distribution

Bayesian networks: example

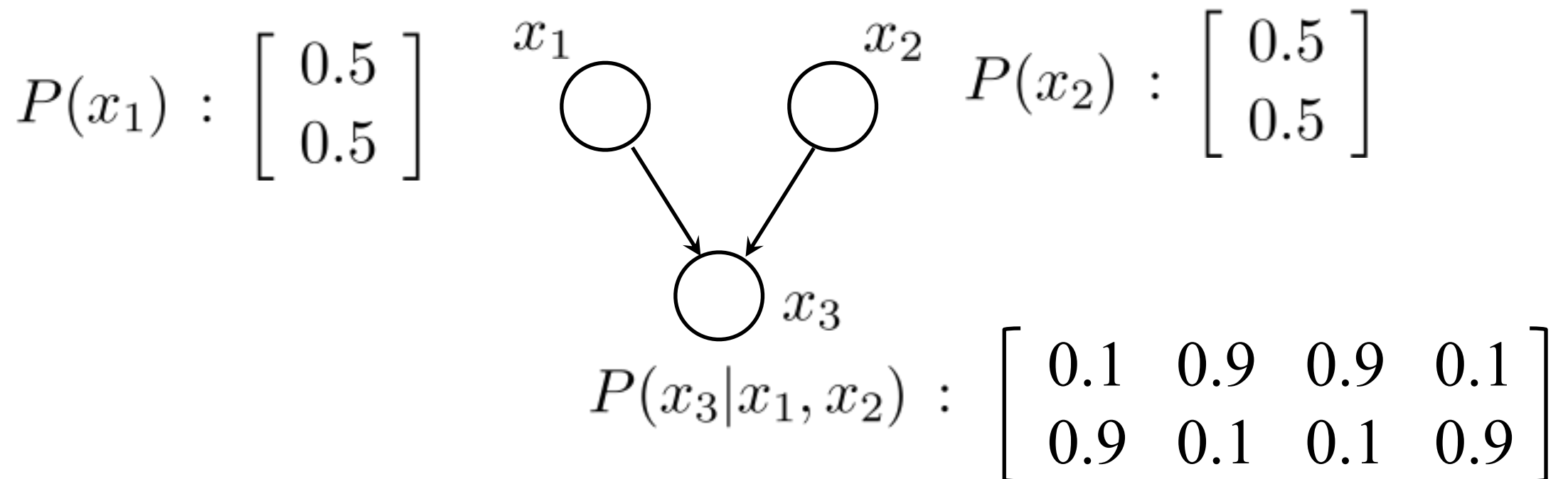


$$P(x_1, x_2, x_3) = P(x_1)P(x_2)P(x_3|x_1, x_2)$$

- The graph alone would permit us to infer two qualitative properties about the underlying distribution
 - x_1 and x_2 are **independent**, for brevity we say $X_1 \perp X_2$ holds

Lecture 1 slide 20:
A and B are independent:
 $P(A, B) = P(A) P(B)$

Bayesian networks: example

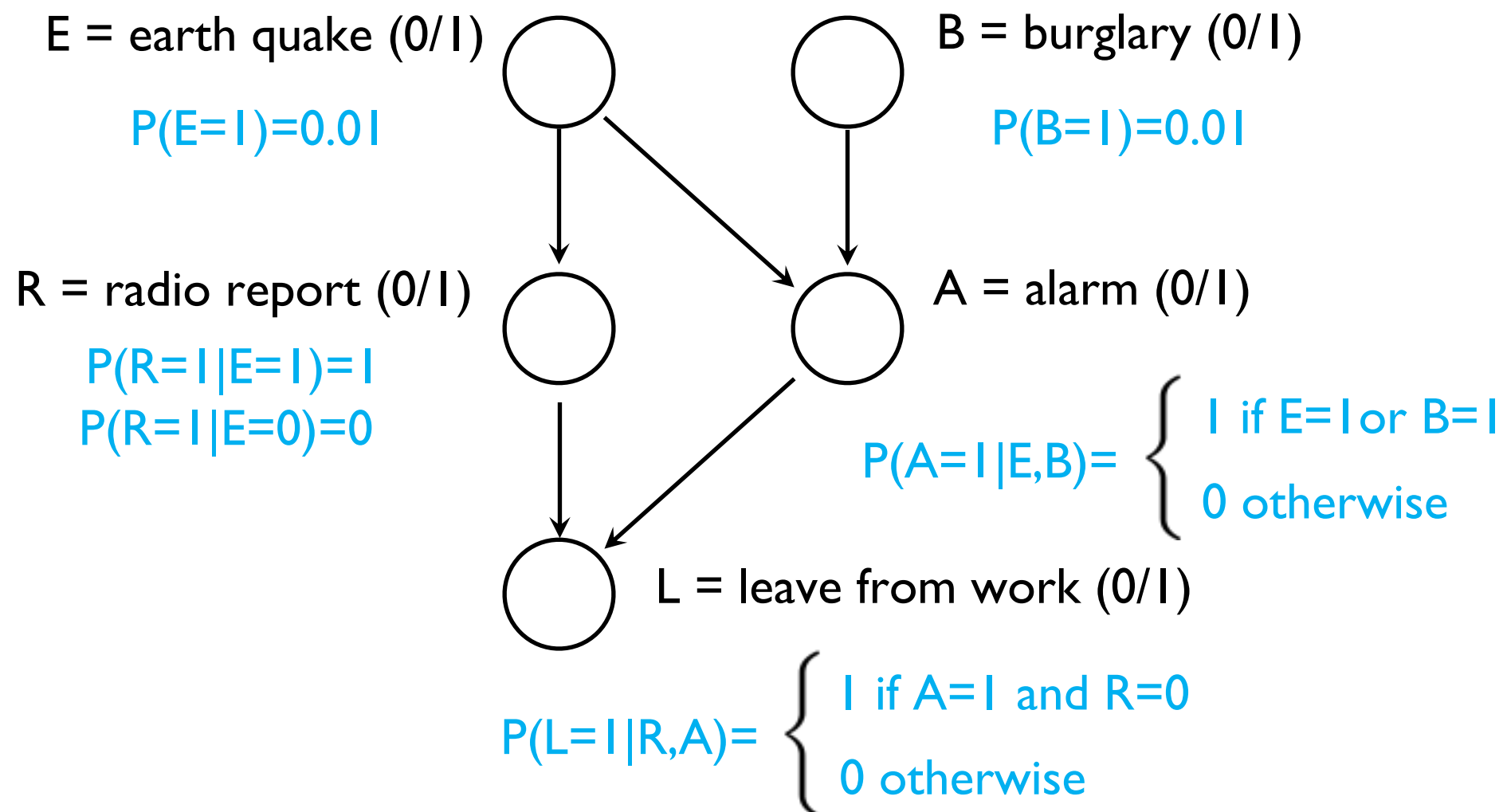


$$P(x_1, x_2, x_3) = P(x_1)P(x_2)P(x_3|x_1, x_2)$$

- The graph alone would permit us to infer two qualitative properties about the underlying distribution
 - x_1 and x_2 are **independent**, for brevity we say $X_1 \perp X_2$ holds
 - x_1 and x_2 are **not conditionally independent** given x_3 , for brevity we say $X_1 \perp X_2 | X_3$ does **not** hold

A and B are conditionally independent given C:
 $P(A, B | C) = P(A | C) P(B | C)$

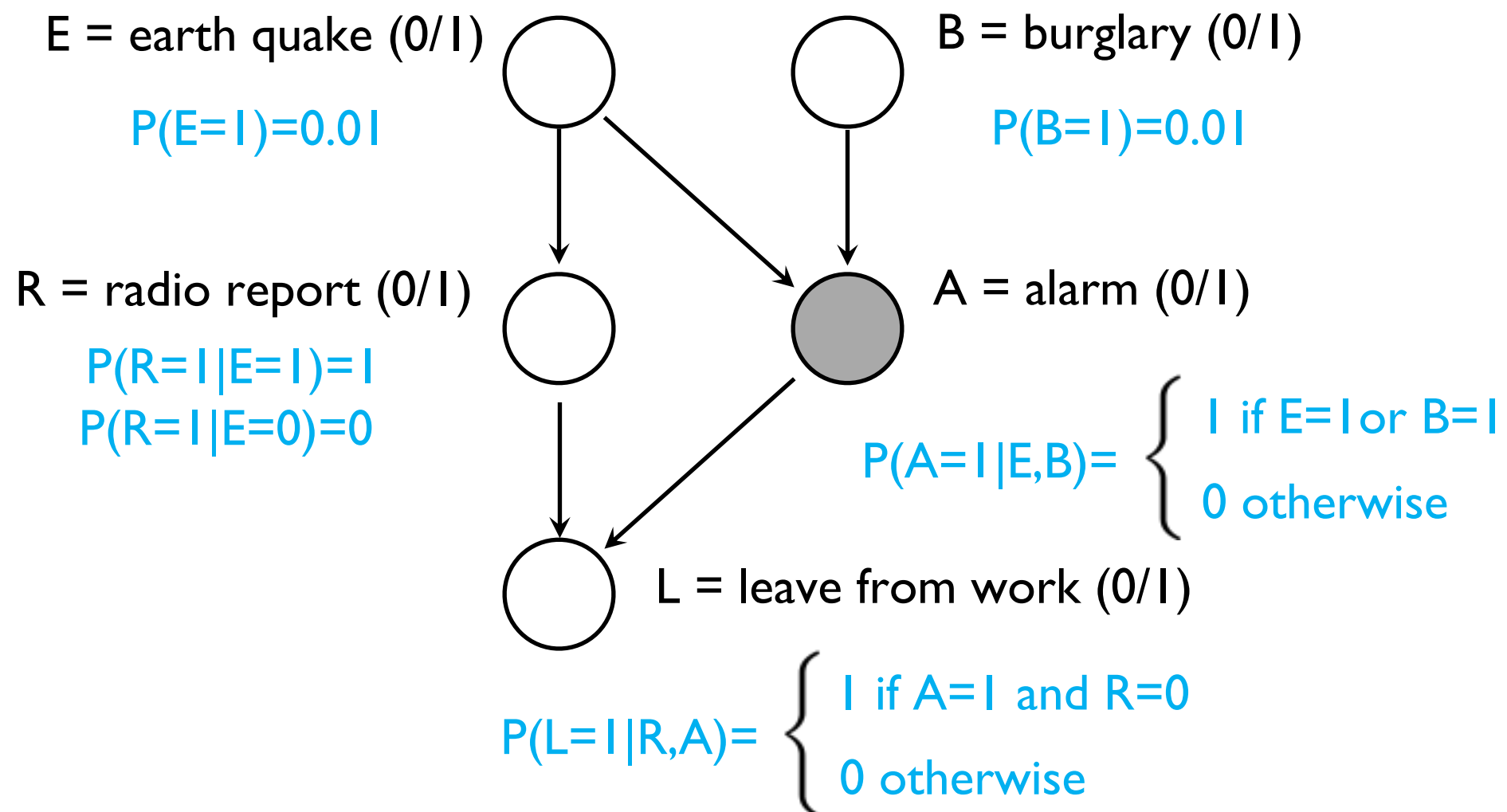
Independence Example 1



Joint distribution:

$$P(E,B,R,A,L) = P(E) P(B) P(R | E) P(A | E, B) P(L | R, A)$$

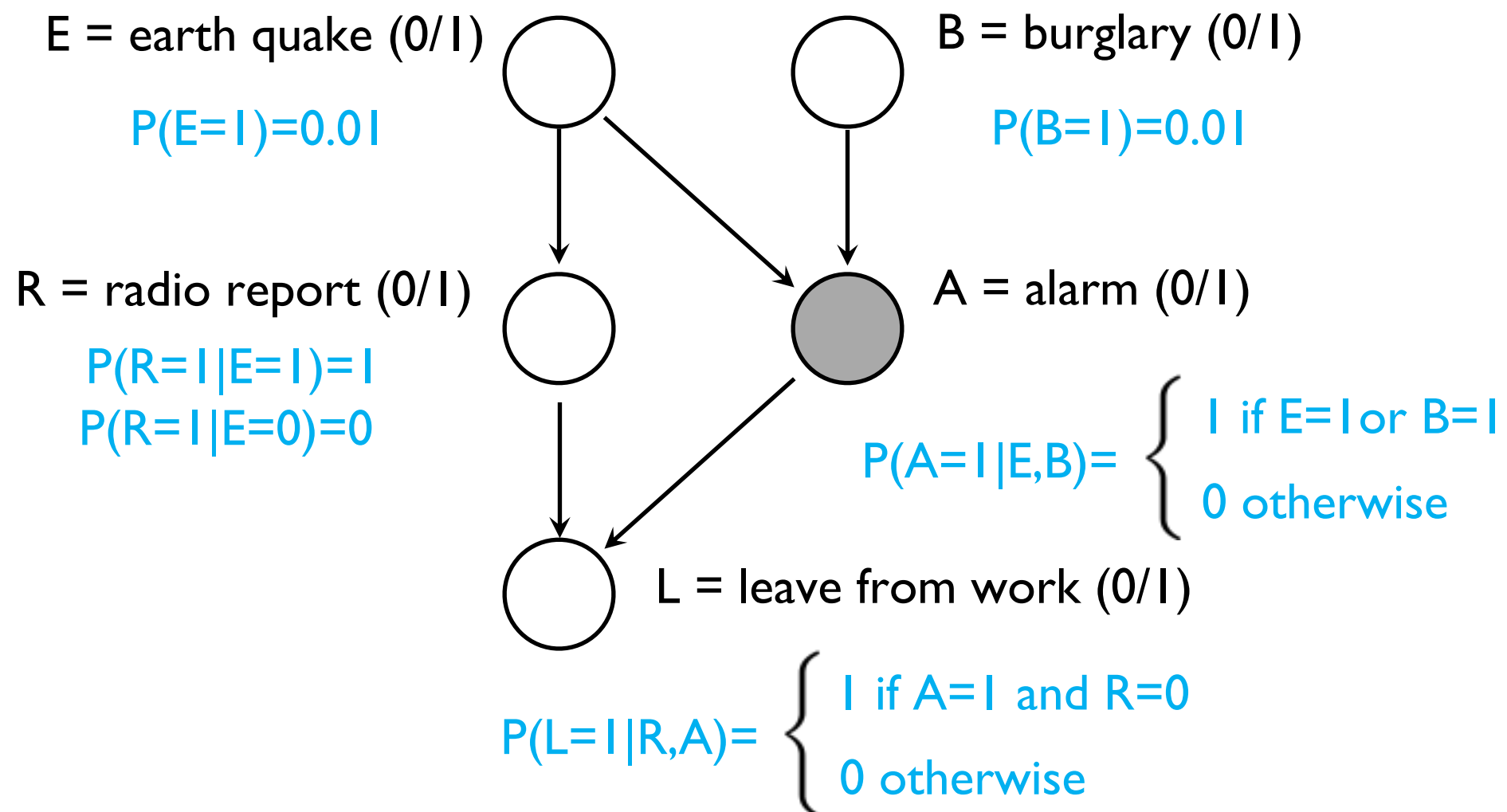
Independence Example 1



- If we hear that the alarm is on, is my leaving from work in any way related to whether there's a burglary?

A bit more generally, we are asking whether L is independent from B given A , i.e., is $L \perp B|A$?

Independence Example 1



- We can know whether $L \perp B | A$ by the formulas on slide 13:
 $P(L, B | A) = P(L | A) P(B | A)$ for all values of L, B, A .
- Fortunately, instead of the above, we only need to use the graph (and not the probabilities.)

Independence and D-separation

- Independence statements in Bayesian networks are encoded in a “graph separation” criterion known as D-separation

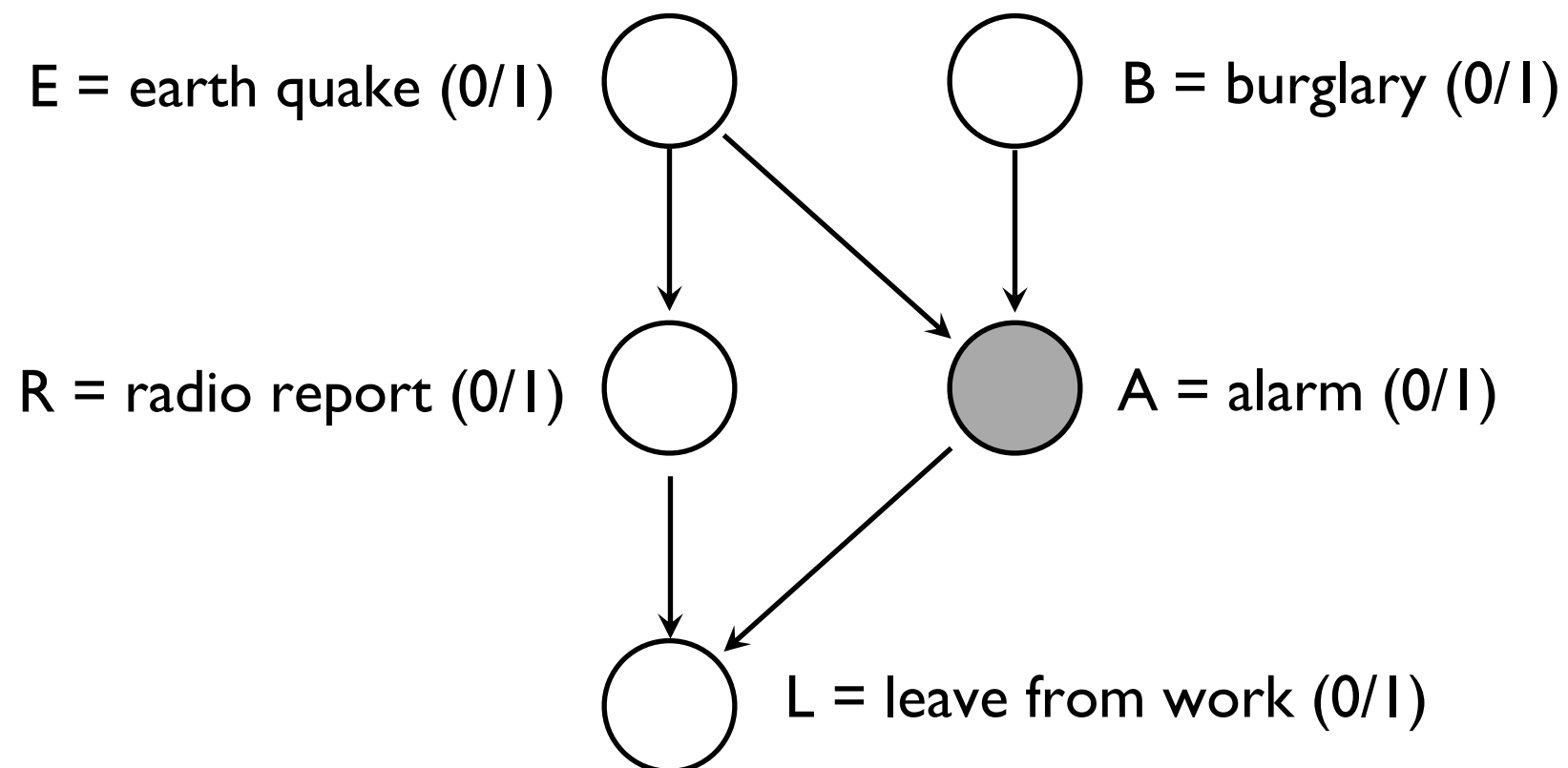
Sets of variables X and Y are independent given S , i.e., $X \perp Y | S$ if they are D-separated in the graph

- Definition: sets of nodes X and Y are D-separated given S if they are separated (via ordinary graph separation) in the moralized ancestral graph.

Ancestral graph: include X and Y and S as well as all of their ancestors (parents, parents parents, etc.)

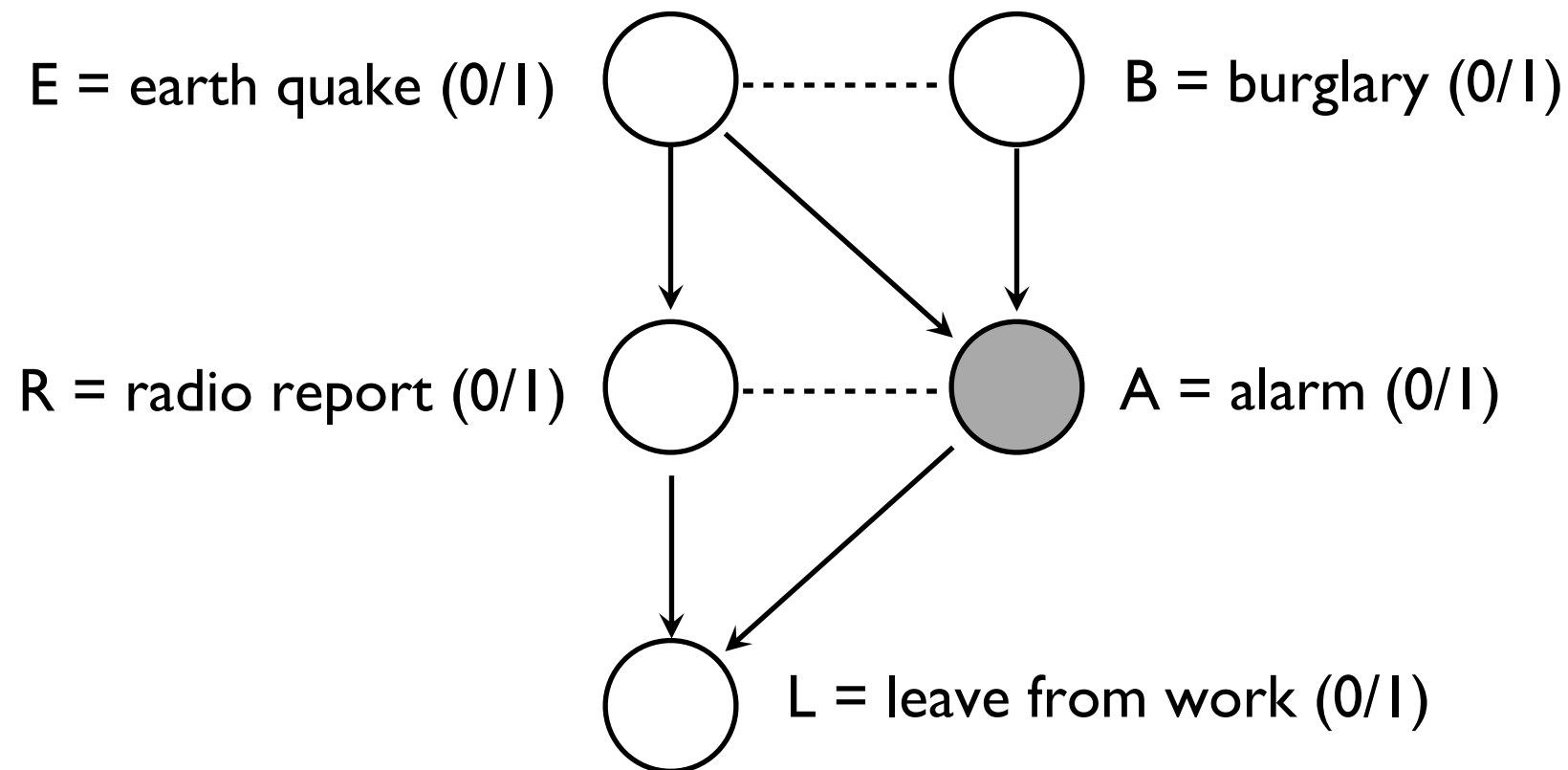
Moral graph: connect (“marry”) each pair of parents

Example 1: Ancestral graph



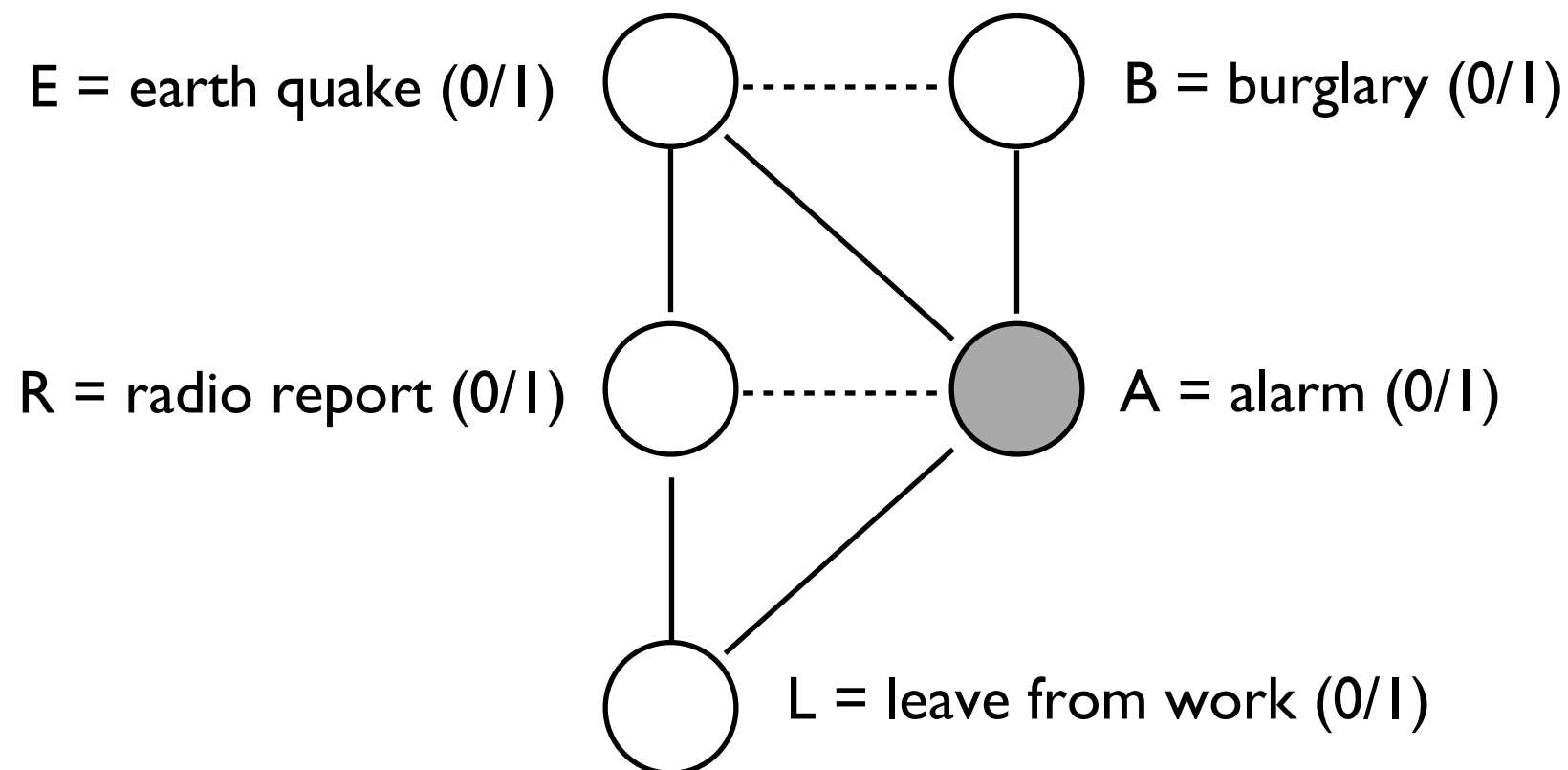
- R and E are both ancestors of L and L , A , and B are included in any case. Thus the ancestral graph is the same as the original graph.

Example 1: Moralized ancestral graph



- The parents of each node are connected (“married”) pairwise, giving rise to the moralized ancestral graph

Example 1: Ordinary graph separation

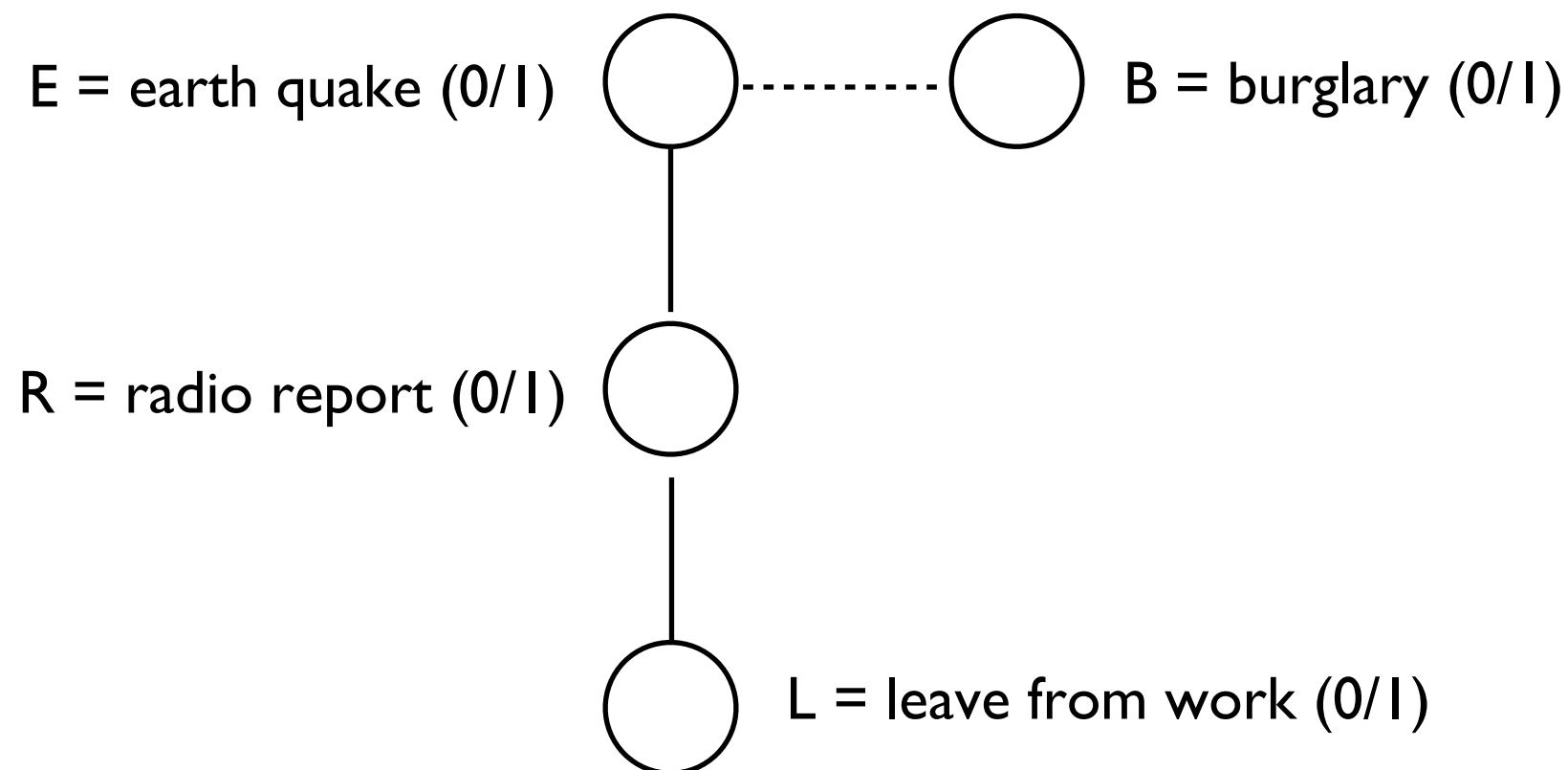


- After turning directed edges into undirected edges, we ask whether L is separated from B given A (it is not!)

$L \perp B | A$ does NOT hold

We can infer that L and B are *dependent* given A

Example 1: Ordinary graph separation

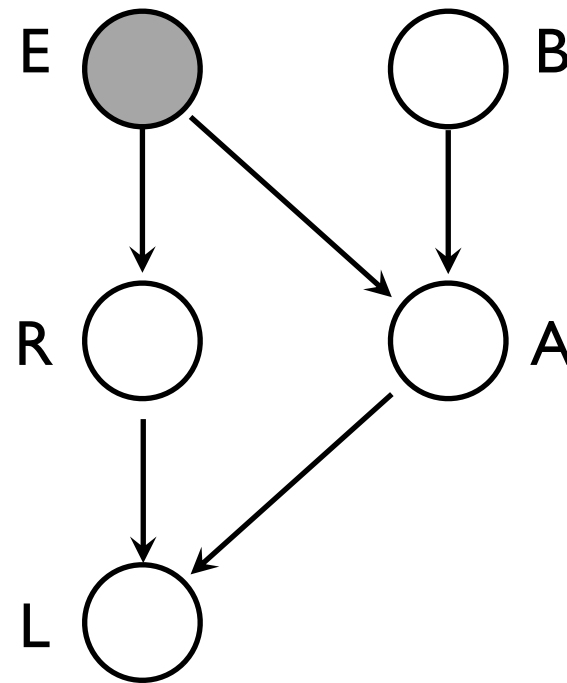


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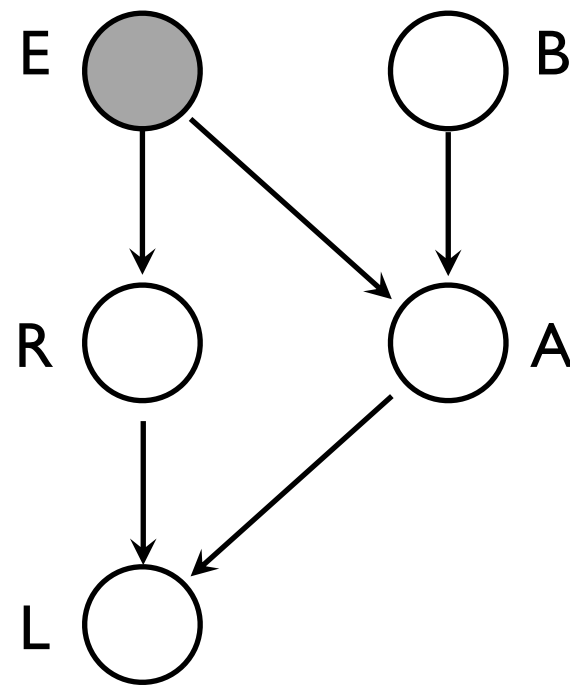
We can infer that L and B are *dependent* given A

Independence Example 2

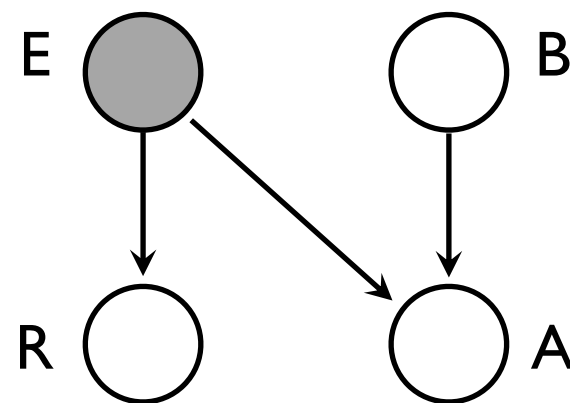


Are R and A independent of each other given E ?

Example 2: Ancestral graph

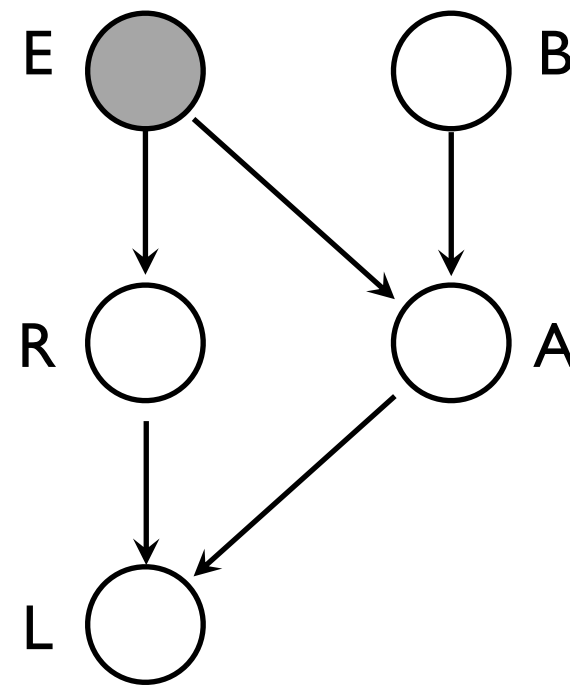


Are R and A independent of each other given E ?

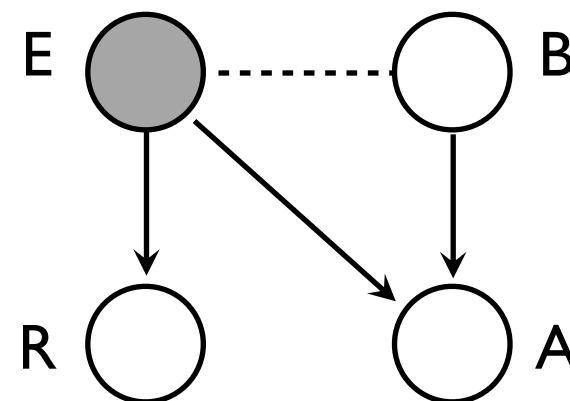


Ancestral graph for R , A and E

Example 2: Moralized ancestral graph

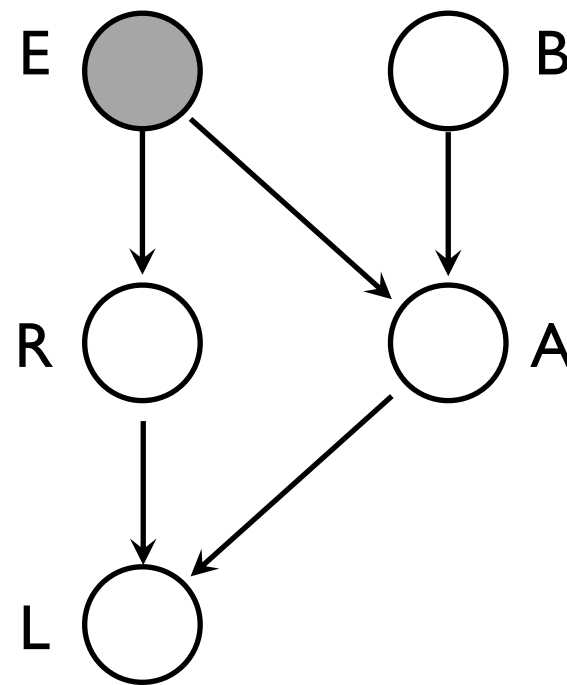


Are R and A independent of each other given E ?

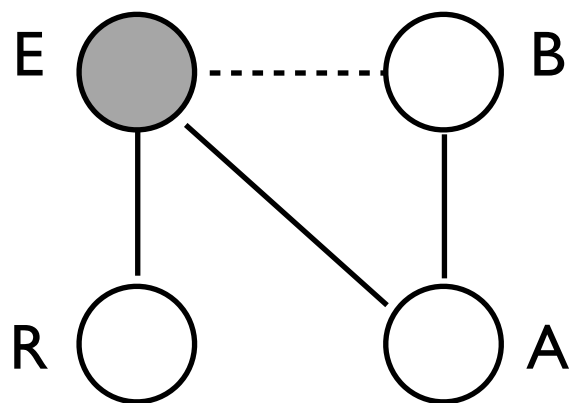


Moralized ancestral graph
for R , A and E

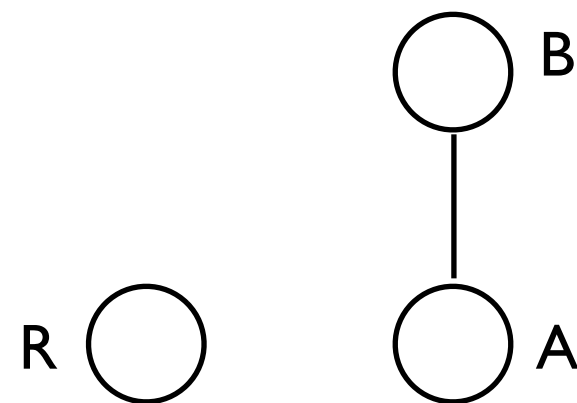
Example 2: Ordinary graph separation



Are R and A independent of each other given E ?

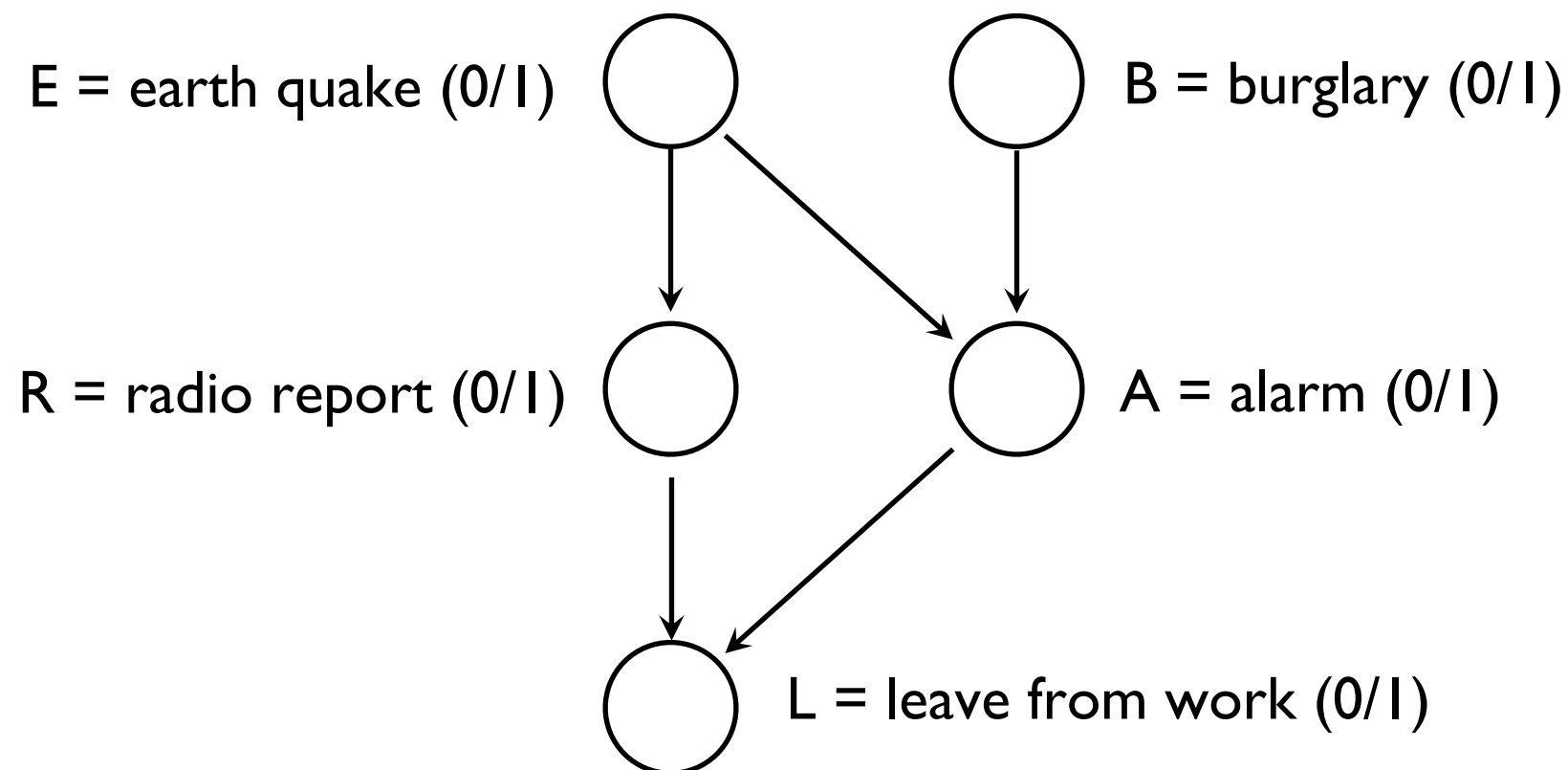


Ordinary graph
separation



R and A are separated given E
 $\Rightarrow R \perp A | E$

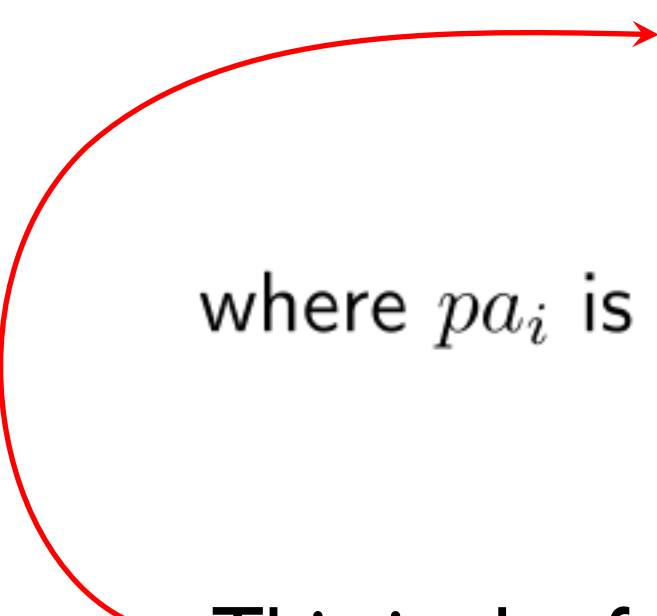
Graphs and probability distributions



- Each graph encodes a large number of conditional independence statements (via D-separation criterion)
- How do we ensure that the distribution we use is consistent with all such statements?

Graphs and probability distributions

- Any distribution $P(x_1, \dots, x_n)$ consistent with all the independence statements implied by an acyclic directed graph (via D-separation) can be written as a product of local conditional distributions of a variable given its parents


$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_{pa_i})$$

See slides 11 and 14 for examples

where pa_i is the set of parents of i in the graph and

$$x_{pa_i} = \{x_j, j \in pa_i\}$$

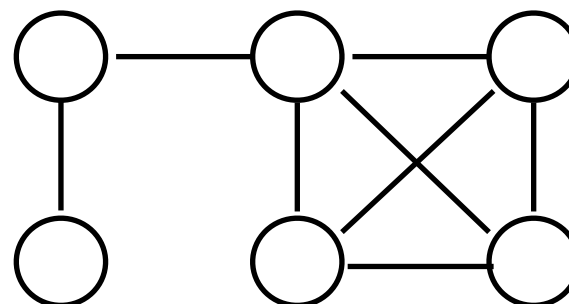
- This is the formal definition of “Bayesian network”

Markov random fields

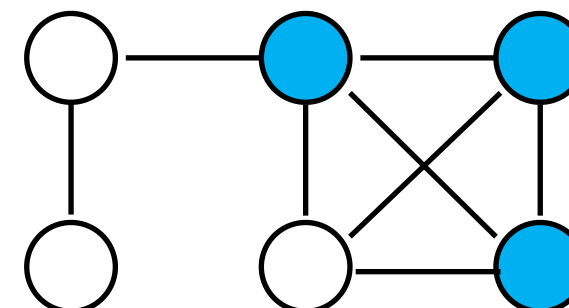
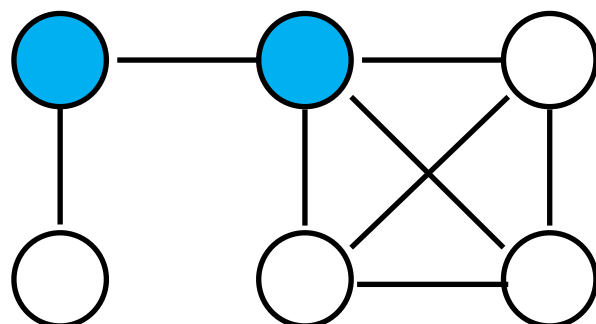
Independence and mapping Bayesian
networks to Markov random fields

Undirected graphs

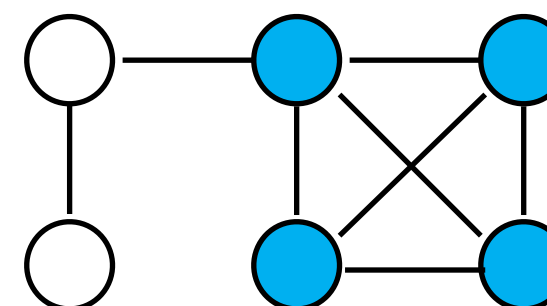
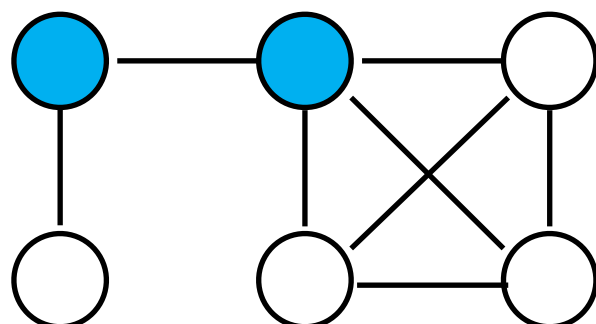
- An undirected graph contains nodes (circles) and undirected edges (lines from one node to another node)



- Clique: set of nodes S for which there is an edge between every two nodes in S



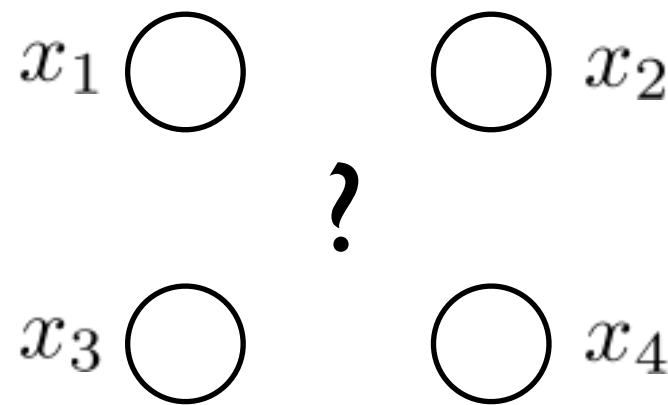
- Maximal clique: a clique that is not included in another clique



Why not Bayesian networks?

- What is the Bayesian network that encodes the following two independence statements and no others?

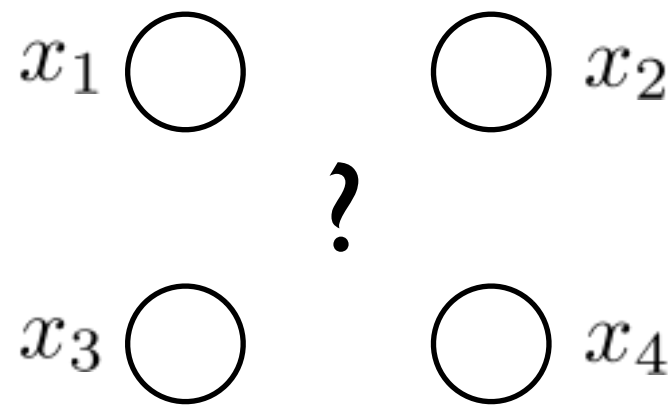
$$X_1 \perp X_4 | X_2, X_3 \quad \text{and} \quad X_2 \perp X_3 | X_1, X_4$$



Why not Bayesian networks?

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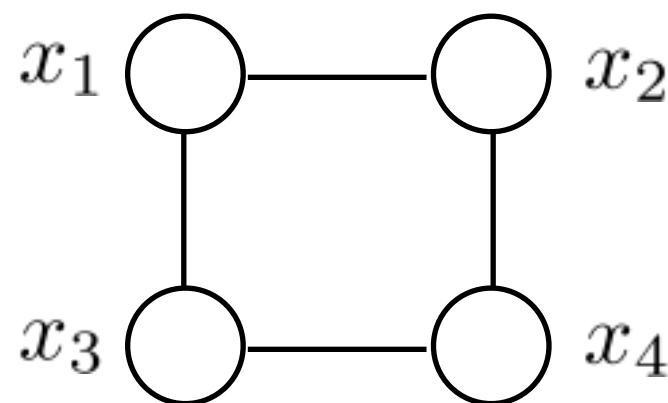
$$X_1 \perp X_4 | X_2, X_3 \quad \text{and} \quad X_2 \perp X_3 | X_1, X_4$$



No such Bayesian network exists !

Undirected graphical models

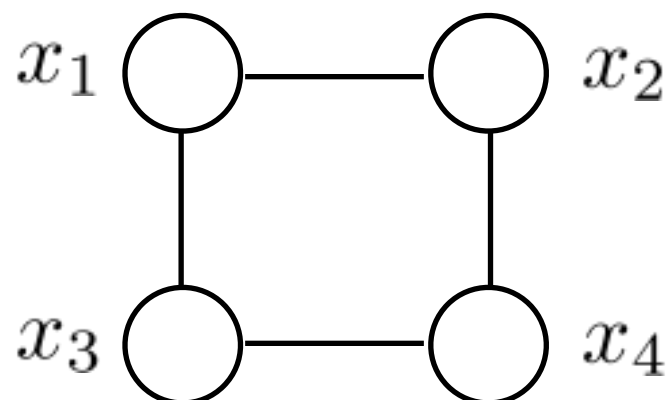
- Markov Random Fields (MRFs) are undirected graphical models where nodes correspond to variables and undirected edges indicate dependencies
- The graph semantics is simpler: X and Y are conditionally independent given S if they are separated in the graph by S (all paths between X and Y go via S)



$$X_1 \perp X_4 | X_2, X_3 \quad \text{and} \quad X_2 \perp X_3 | X_1, X_4$$

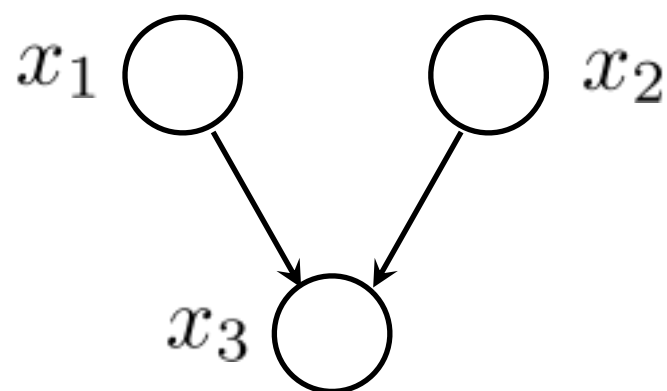
Comparison

- As graphical models, MRFs are complementary to Bayesian networks



There is no Bayesian network that implies only these independencies

$$X_1 \perp X_4 | X_2, X_3 \quad \text{and} \quad X_2 \perp X_3 | X_1, X_4$$

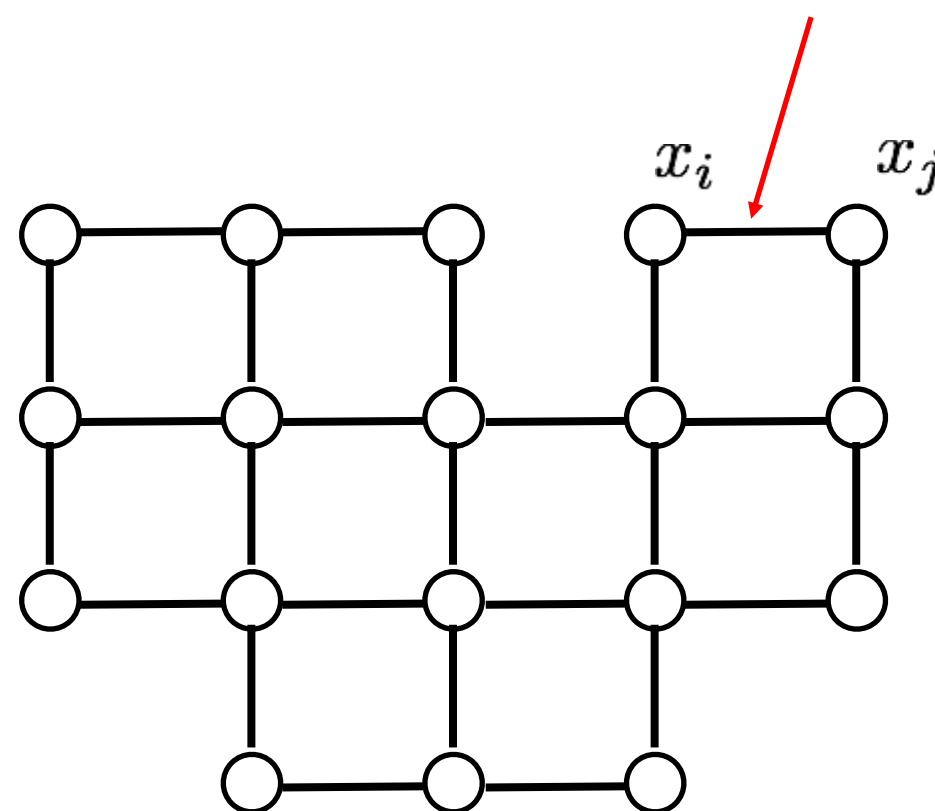


There is no MRF that implies only this marginal independence

$$X_1 \perp X_2$$

Simple example

- Suppose we model each room as a node in a graph such that adjacent rooms are linked by an edge
- We introduce a discrete variable $x_i \in \{-1, 1\}$ for each node representing whether the temperature in the room is hot (1) or cold (-1)
- The temperatures between any two adjacent rooms tend to depend on each other so we couple the temperatures through real valued “potential functions” $\theta_{ij}(x_i, x_j)$



θ_{ij} receives two $+1/-1$ parameters and returns a scalar, e.g.,

Let $x_i = -1, x_j = 1$
 $\theta_{ij}(x_i, x_j) = \theta_{ij}(-1, 1) = -5.7$

Let $x_i = 1, x_j = 1$
 $\theta_{ij}(x_i, x_j) = \theta_{ij}(1, 1) = 2.9$

The probability model

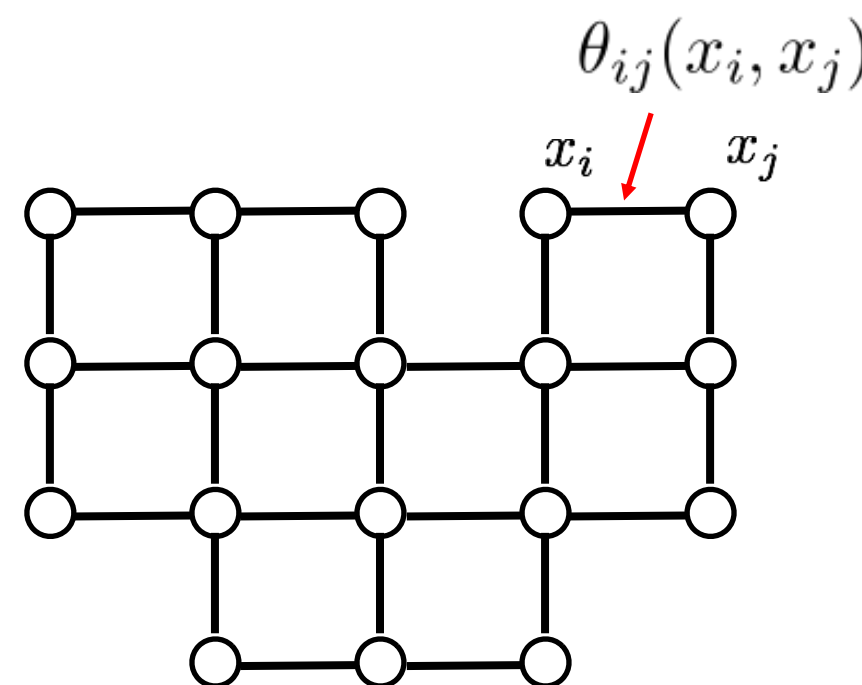
- The couplings on the edges $(i, j) \in E$ between adjacent rooms give rise to a joint probability distribution over all the temperatures

$$P(x_1, \dots, x_n; \theta) = \frac{1}{Z(\theta)} \exp \left(\sum_{(i,j) \in E} \theta_{ij}(x_i, x_j) \right)$$

where the “partition function”

$$Z(\theta) = \sum_{x_1, \dots, x_n} \exp \left(\sum_{(i,j) \in E} \theta_{ij}(x_i, x_j) \right)$$

normalizes the distribution (globally).

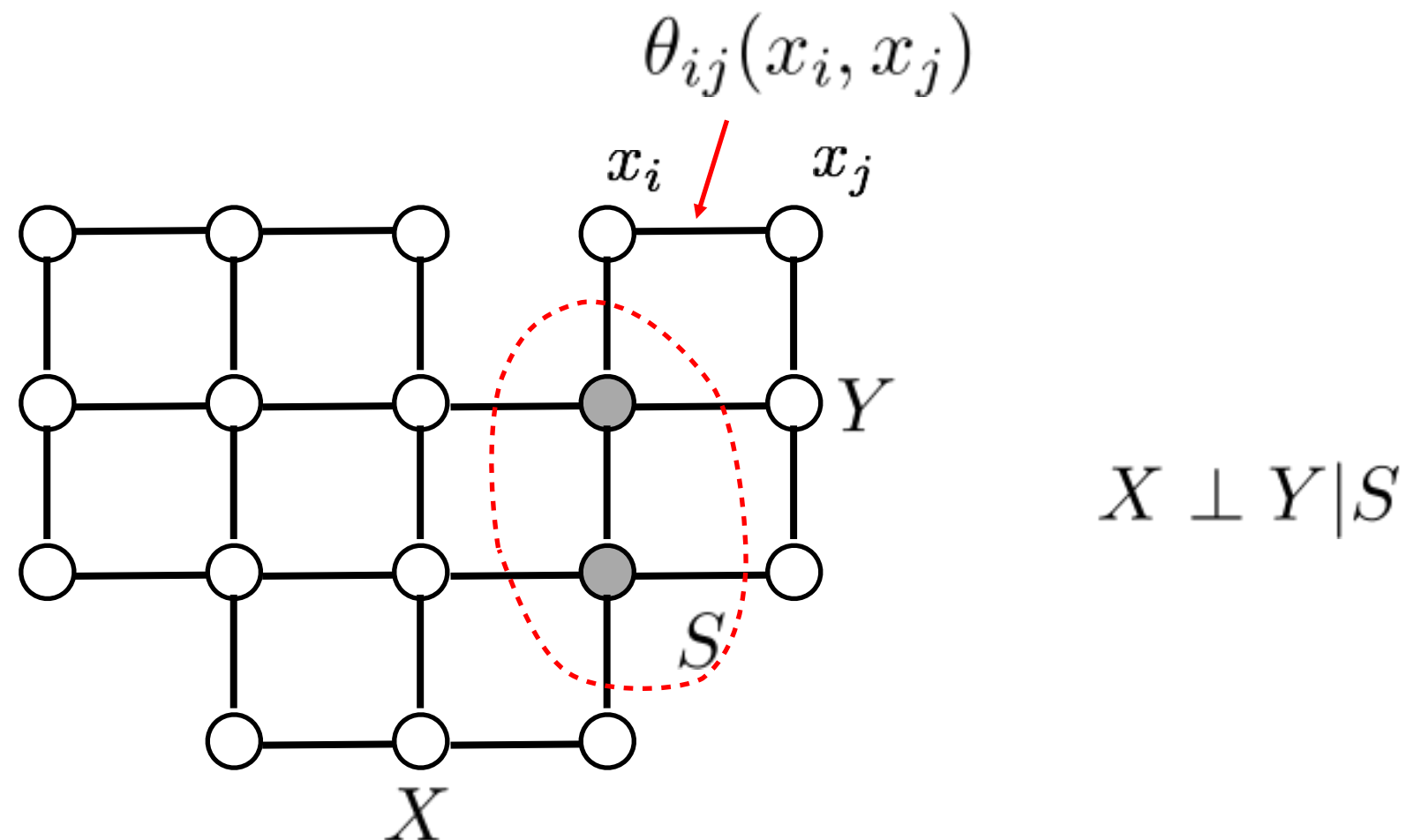


The probability model

- The joint probability distribution

$$P(x_1, \dots, x_n; \theta) = \frac{1}{Z(\theta)} \exp \left(\sum_{(i,j) \in E} \theta_{ij}(x_i, x_j) \right)$$

is consistent with all the independence statements implied by the undirected graph (and the simple graph separation criterion)



If we remove nodes in S , and X and Y get separated, then X and Y are independent given S

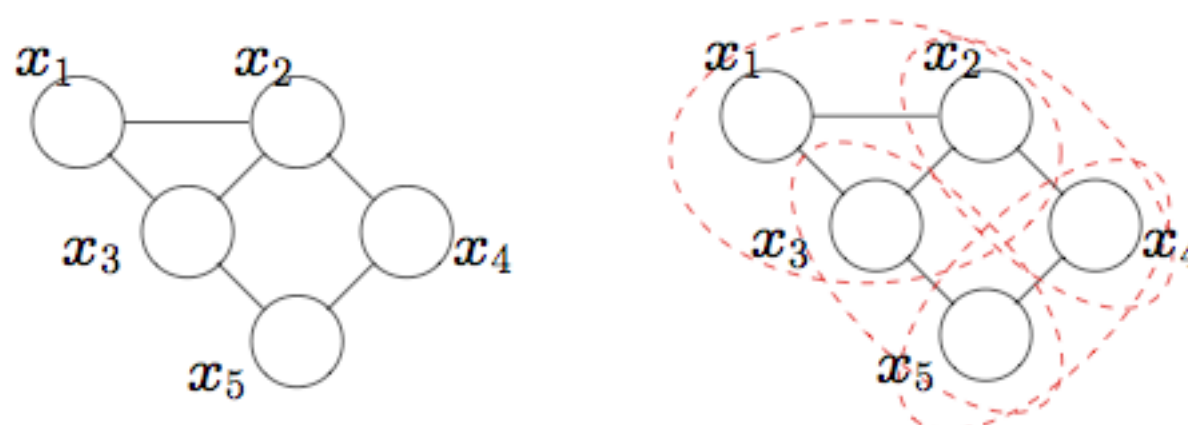
Distribution, consistency

- The simple graph separation criterion places constraints on the distributions we can associate with an undirected graph

Theorem: (Hammersley-Clifford) Any distribution that is consistent with an undirected graph has to factor according to the (maximal) cliques in the graph

$$P(x_1, \dots, x_n) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \psi_c(x_c)$$

where $x_c = \{x_j, j \in c\}$ denotes the variables in clique c . $\psi_c(x_c)$ is any positive valued function of the variables contained in clique c (clique potential).



Example cont'd

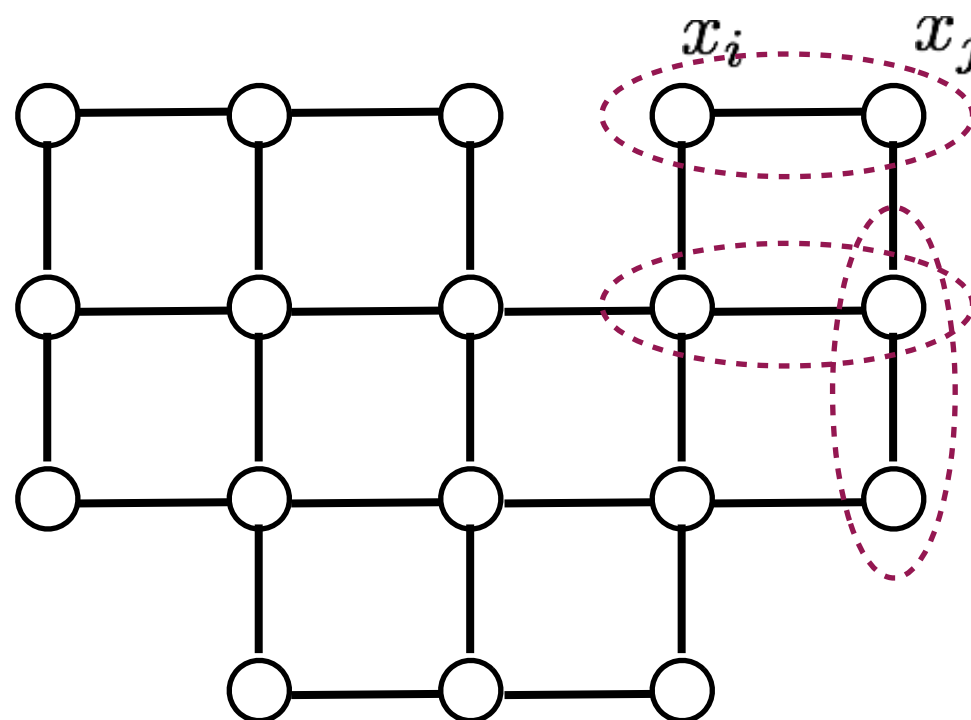
- In the example, each clique corresponds to an edge in the graph and the distribution factors according to the edges

$$P(x_1, \dots, x_n; \theta) = \frac{1}{Z(\theta)} \exp \left(\sum_{(i,j) \in E} \theta_{ij}(x_i, x_j) \right)$$

$$= \frac{1}{Z(\theta)} \prod_{(i,j) \in E} \exp \left(\theta_{ij}(x_i, x_j) \right)$$

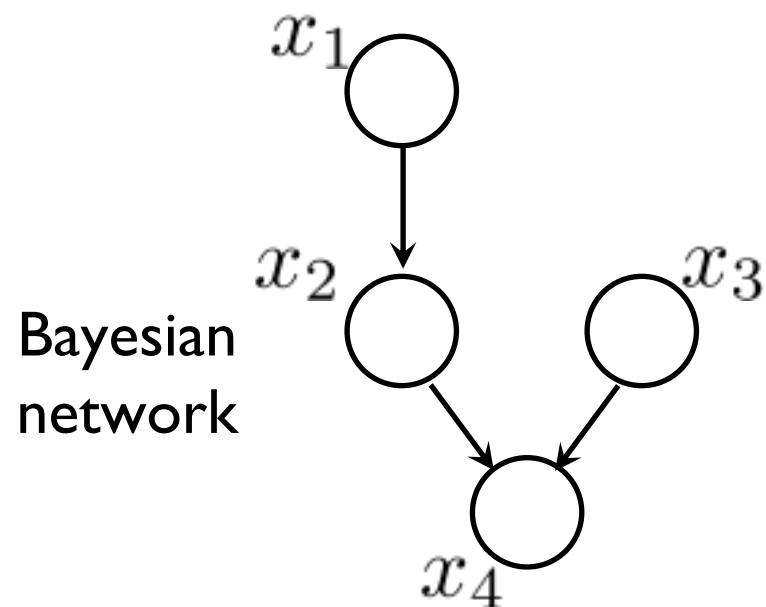
$\psi_{ij}(x_i, x_j)$

positive valued function
of the variables assoc.
with the edge



From Bayesian networks to MRFs

- We can turn any Bayesian network into a MRF model in two steps (potentially loosing some of the independence properties that were explicit in the directed graph)

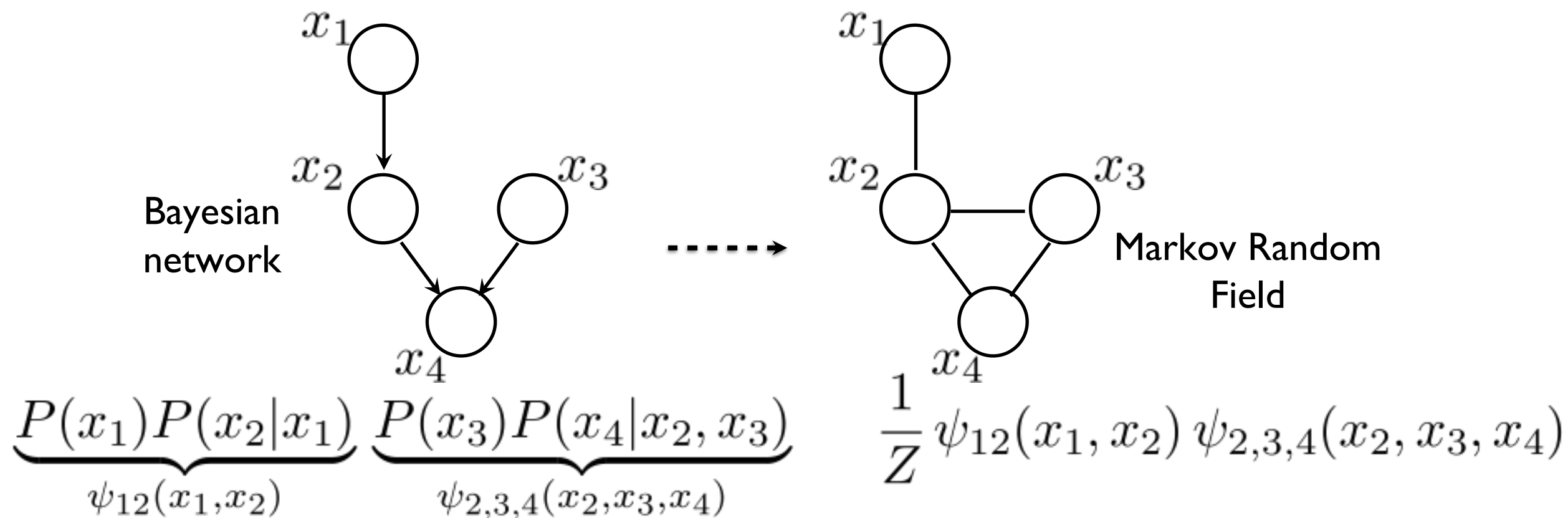


$$P(x_1)P(x_2|x_1)P(x_3)P(x_4|x_2, x_3)$$

From Bayesian networks to MRFs

- We can turn any Bayesian network into a MRF model in two steps (potentially loosing some of the independence properties that were explicit in the directed graph)

Graph transformation: moralize and omit edge directions



Probability distribution: map each conditional probability to a clique potential that contains it (mapping is not necessarily unique). $Z = 1$ in this case.

Mini Summary

- Bayesian networks (BNs) represented by directed acyclic graphs
 - * Independence by ancestral graph, moralization, ordinary graph separation
- Markov random fields (MRFs) represented by undirected graphs
 - * Just graph separation
- Can transform BNs into MRFs

Next: Inference on MRFs